



Bansilal Ramnath Agarwal Charitable Trust's
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Subject Name & Code: Cloud and DevOps

Assignment No: 05

Title of Assignment: Containerization with Docker – Install Docker, create containers, manage images, and use volumes

Date of Performance: / /2025

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1. Introduction

Containerization is a lightweight form of virtualization where applications and their dependencies are packaged together in isolated units called **containers**. Unlike virtual machines, containers share the host system's operating system kernel, making them more efficient and portable.

Docker is the most widely used containerization platform. It simplifies application deployment by allowing developers to build once and run anywhere.

This assignment explains how to install Docker, create containers, manage images, and use volumes.

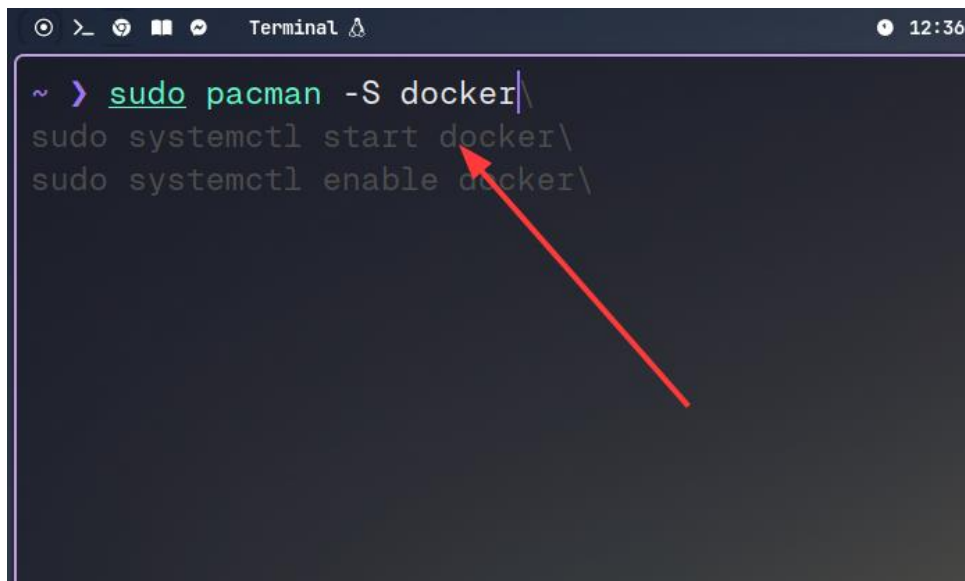
2. Objectives

1. To understand the concept of containerization and Docker.
2. To install and configure Docker on a system.
3. To create and run containers.
4. To manage Docker images effectively.
5. To use Docker volumes for persistent storage.

3. Procedure

Step 1: Install Docker

1.

A terminal window titled "Terminal" with a clock showing 12:36. The prompt is "~ >". The user has entered the command "sudo pacman -S docker\". Below this, two additional commands are listed: "sudo systemctl start docker\" and "sudo systemctl enable docker\". A red arrow points from the bottom right towards the second command.

```
~ > sudo pacman -S docker\  
sudo systemctl start docker\  
sudo systemctl enable docker\
```

2. **Step II:** Verify Installation

```
sudo systemctl enable docker\
> docker --version
Docker version 28.3.3, build 980b856816
~ > |
```

3. Step II: Enable Docker

```
> sudo systemctl start docker
sudo systemctl enable docker

[sudo] password for onkar:
~ > |
```

and check
default)

4: Enable Docker
service (disable

```
~
~
lines 1-23/23 (END)...skipping...
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
   Active: active (running) since Sat 2025-09-13 23:41:14 IST; 58min ago
 Invocation: 54199f9d4c1c43519de2c74145b630de
  TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 961 (dockerd)
      Tasks: 24
    Memory: 90.6M (peak: 92.6M)
       CPU: 1.381s
    CGroup: /system.slice/docker.service
            └─961 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Sep 13 23:41:14 archlinux dockerd[961]: time="2025-09-13T23:41:14.185491544+05:30" level=warning msg="Error (Una>
Sep 13 23:41:14 archlinux dockerd[961]: time="2025-09-13T23:41:14.325103051+05:30" level=warning msg="error loca>
Sep 13 23:41:14 archlinux dockerd[961]: time="2025-09-13T23:41:14.325582859+05:30" level=info msg="Loading conta>
Sep 13 23:41:14 archlinux dockerd[961]: time="2025-09-13T23:41:14.445417816+05:30" level=warning msg="Not using >
Sep 13 23:41:14 archlinux dockerd[961]: time="2025-09-13T23:41:14.445585619+05:30" level=info msg="Docker daemon>
Sep 13 23:41:14 archlinux dockerd[961]: time="2025-09-13T23:41:14.446034742+05:30" level=info msg="Initializin>
Sep 13 23:41:14 archlinux dockerd[961]: time="2025-09-13T23:41:14.495582273+05:30" level=info msg="Completed bui>
Sep 13 23:41:14 archlinux dockerd[961]: time="2025-09-13T23:41:14.509698986+05:30" level=info msg="Daemon has co>
Sep 13 23:41:14 archlinux dockerd[961]: time="2025-09-13T23:41:14.509844704+05:30" level=info msg="API listen on>
Sep 13 23:41:14 archlinux systemd[1]: Started Docker Application Container Engine.
~
~
~
```

Step :Pulling Kali linux docker container:

```
> docker pull kalilinux/kali-rolling
Using default tag: latest
latest: Pulling from kalilinux/kali-rolling
Digest: sha256:698a5d17193a49552200023317bec782c90395ad361f65dda4ee25a45d1f3a06
Status: Image is up to date for kalilinux/kali-rolling:latest
docker.io/kalilinux/kali-rolling:latest
~ > █
```

Step: Checking out running docker containers:

```
> docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
c0a029e9483c   mongo:4.4  "docker-entrypoint.s..." 7 weeks ago    Up 3 seconds  0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp
~ > █
```

Running Kali as a container:

```
(root@6900d8578d08)-[/]
# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var

(root@6900d8578d08)-[/]
# whoami
root

(root@6900d8578d08)-[/]
# █
```

Conclusion:

This assignment covered the fundamentals of Docker containerization, including installation, container creation, image management, and persistent storage with volumes. Docker makes application deployment faster, more reliable, and portable across environments. Mastery of Docker is essential for modern DevOps, cloud computing, and microservices architecture.